

Campaign Level

Budget pacing determines spending pressure



Campaign state

- Remaining budget
- Current pacing reference
- Time and delivery status



Campaign control

(e.g., traffic-aware, PID, dual)

a_{it}^{ref}



Opportunity Level

Autobidder evaluates the current impression



Impression features

- User and context
- Ad and creative features
- Expected value / conversion



Autobidder

(e.g., learned bidder)

a_{it}^{raw}



Execution boundary κ

$$a_{it}^{exec} = clip(a_{it}^{raw}, \max\{0, (1 - \kappa)a_{it}^{ref}\}, (1 + \kappa)a_{it}^{ref})$$



Auction Environment

Many advertisers compete for each impression

Advertiser 1

Advertiser 2

...

Advertiser N



Real-time auction

(e.g., second price)

a_{it}^{exec}



Winner



1. Value-Delivery Frontier

How does execution discretion affect campaign outcomes?



Campaign

(state, goals, targeting)

Execution width κ



Performance

(e.g., conversions, CPA, ROAS)



Delivery

(e.g., spend rate, budget delivery)



2. Drivers of Discretion Value

When does wider discretion help?



Incremental selection information



Campaign selection room



Campaign load



3. Platform Deployment and Competitive Spillovers

What happens when many advertisers adopt execution discretion?

Advertiser 1

Advertiser 2

...

Advertiser N



Shared marketplace

(real-time auction)



Ad allocation



Clearing prices



Platform revenue



Spillovers